Data Analytics Capstone

Data-Driven Analysis and Predictive Modeling of Graduate Employment and Salary Outcomes in Singapore

Synopsis

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QM640: Data Analytics Capstone

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Term 3

July 12, 2026

Introduction

Background and Context

Graduate employment outcomes are a key indicator of how well higher education connects with labor-market needs. Students and families use employment rates and starting salary information when choosing degree programs, while universities and policymakers use such outcomes to assess program relevance, workforce alignment, and skills development. In Singapore, these questions are important because university education is closely linked to national competitiveness and employability planning.

The Singapore Graduate Employment Survey (GES), published by the Ministry of Education (MOE) through data.gov.sg, provides program-level employment and salary outcomes for graduates from the autonomous universities. The dataset includes information such as university, school, degree, overall employment rate, full-time permanent employment rate, and salary indicators collected around six months after final examinations (Ministry of Education, 2026).

This study moves beyond simple ranking by salary or employment rate. A degree program may show high overall employment but lower full-time permanent employment, or high median salary but greater salary spread and year-to-year volatility. A data analytics framework can therefore provide a more balanced view by combining descriptive analytics, inferential statistics, regression, machine learning, and clustering.

Problem Statement

Graduate outcome data are often interpreted through isolated indicators such as median gross monthly salary or overall employment rate. This can create an incomplete picture because graduate career outcomes are multidimensional and may vary by university, degree, year, employment stability, salary distribution, and outcome consistency. The gap addressed in this study is the absence of a concise analytical framework that jointly explains, predicts, and segments Singapore graduate employment and salary outcomes using official program-level data.

Literature Gap and Contribution

Existing literature discusses graduate employability as a multidimensional concept and also shows that machine learning can be applied to employability prediction. However, the official Singapore GES data is often interpreted using separate indicators such as salary or employment rate. This study addresses that gap by combining employment, salary, stability, regression, prediction, and clustering into one program-level decision-support framework.

Purpose of the Study

The purpose of this study is to develop a data-driven analytical framework for evaluating graduate employment and salary outcomes in Singapore. The study will analyse employment and salary trends, examine differences across universities and degree programs, identify the key drivers of graduate outcomes, develop predictive models for high-performing programs, and segment programs into meaningful employability profiles. A Graduate Career Value Index (GCVI) will be used as a constructed decision-support measure that combines employment, salary, and stability indicators.

Scope and Objectives

The unit of analysis is a university-degree-year observation. The study is limited to program-level GES records and will not make claims about individual student employability because the dataset does not contain individual background, grades, internships, or job-search behavior.

- Summarize employment and salary trends across Singapore university degree programs.
- Test whether employment and salary outcomes differ across universities and degree programs.
- **I**dentify the key factors associated with graduate salary outcomes and career value.
- Build predictive models for high-employment, high-salary, and high-career-value programs.
- Segment programs into employability-salary profiles for decision support.

Link Between Research Gap and Research Questions

Existing graduate outcome reporting often presents employment and salary indicators separately, which may not fully capture the multidimensional nature of graduate career outcomes. This study addresses that gap by examining trends over time, testing differences across universities and degree programs, identifying salary and career-value drivers, predicting high-outcome programs, and grouping programs into meaningful employability-salary profiles. Therefore, the research questions are structured to move from descriptive analysis to diagnostic comparison, explanatory modelling, predictive analytics, and segmentation-based decision support.

Research Questions (RQ)

RQ1: How have employment and salary outcomes evolved across Singapore university degree programs from 2013 to 2025?

RQ2: Do employment and salary outcomes differ significantly across universities and degree programs?

RQ3: Which factors contribute most to graduate salary outcomes and graduate career value across Singapore university programs?

RQ4: Can machine learning models accurately predict high-employment, high-salary, and high-career-value degree programs?

RQ5: Can degree programs be segmented into meaningful employability-salary profiles to support student, university, and policy decision-making?

Sample Size Calculation

The available dataset contains **1,550 program-year observations before cleaning**. Although the study uses the full available dataset rather than a sampled subset, sample-size adequacy was assessed to ensure sufficient observations for statistical and machine-learning analyses.

For a 95% confidence level and 5% margin of error, the conservative minimum sample size for proportion-based analysis is calculated as:

$$n = \frac{Z^2 p(1-p)}{e^2}$$

where $Z = 1.96$, $p = 0.5$, and $e = 0.05$. Therefore:

$$n = \frac{1.96^2 \times 0.5 \times 0.5}{0.05^2}$$

$$n = \frac{3.8416 \times 0.25}{0.0025}$$

$$n = 384.16 \approx 385$$

The available dataset of **1,550 observations** exceeds this minimum requirement.

For regression analysis, Green’s rule provides an additional check:

$$N \geq 50 + 8m$$

where m is the number of predictors. Assuming approximately eight predictors:

$$N \geq 50 + 8(8) = 114$$

The available dataset substantially exceeds the minimum requirements for the proposed statistical and predictive analyses. Therefore, the full valid dataset after cleaning will be used. For machine learning, model performance will be evaluated using train-test splitting and cross-validation. For clustering, cluster adequacy will be assessed using silhouette score, stability, and interpretability.

Sample-size table

RQ	Method Used	Key Parameters	Minimum N
RQ1	Confidence interval; trend analysis	Z=1.96, p=0.5, e=0.05	$n \approx 385$; full dataset exceeds requirement
RQ2	ANOVA / Kruskal-Wallis	Group comparisons; post-hoc tests	Full dataset used; group sizes checked after cleaning
RQ3	Regression / feature importance	Green's rule: N >= 50 + 8m; m≈8	Minimum $N = 114$; use larger available sample
RQ4	Classification with cross-validation	Train-test split; F1, ROC-AUC, balanced accuracy	Full dataset used; class representation checked
RQ5	Clustering / PCA	Standardized features; silhouette score	Full dataset used; cluster adequacy evaluated by validation metrics

Data Description

Data Source

The primary dataset is the Graduate Employment Survey - NTU, NUS, SIT, SMU, SUSS & SUTD, published by MOE on data.gov.sg. The dataset provides program-level graduate employment and salary outcomes for Singapore autonomous universities and includes variables related to university, school, degree, employment rates, and monthly salary indicators (Ministry of Education, 2026).

Source URL: https://data.gov.sg/datasets/d_3c55210de27fcccda2ed0c63fdd2b352/view

Figure 1. Data source snapshot from data.gov.sg.

Note: Snapshot recreated from official dataset metadata for source documentation.

Data Dictionary

Variable	**Type**	**Description**	**Use in Analysis**
year	Numeric/categorical	Survey year	Trend, regression, time effects
university	Categorical	University offering the degree	Group comparison, modelling
school	Categorical	Faculty or academic unit	Program grouping
degree	Categorical/text	Degree program name	Program comparison
employment_rate_overall	Numeric	Overall employment percentage	Outcome, GCVI, clustering
employment_rate_ft_perm	Numeric	Full-time permanent employment percentage	Employment stability, GCVI
basic_monthly_median	Numeric	Median basic monthly salary	Salary analysis
gross_monthly_median	Numeric	Median gross monthly salary	Primary salary outcome

gross_monthly_25th/75th	Numeric	Salary quartiles	Salary spread
salary_iqr	Derived	75th percentile minus 25th percentile	Dispersion indicator
salary_premium	Derived	Program salary minus annual median	Relative salary performance
employment_premium	Derived	Program employment minus annual average	Relative employment performance
stability_score	Derived	Lower year-to-year volatility	GCVI and clustering
career_value_index	Derived	Composite employment-salary-stability measure	GCVI output and ML target
high_employment / high_salary / high_career_value	Binary	Threshold-based outcome labels	Classification targets

Data Preparation and Repository

Data preparation will include importing the CSV file, standardizing names, converting salary and employment fields to numeric format, checking missing values, creating derived variables, encoding categorical variables, and standardizing numerical variables for clustering and PCA. The cleaned data and code will be made available through GitHub, with separate folders for raw data, processed data, notebooks, outputs, and dashboard files.

Analytic Approach

The analysis follows a progression from descriptive analytics to diagnostic, predictive, and segmentation analytics. RQ1 describes what happened over time. RQ2 diagnoses whether statistically significant differences exist. RQ3 explains salary and career-value drivers. RQ4 predicts high-outcome programs. RQ5 converts the findings into actionable program profiles.

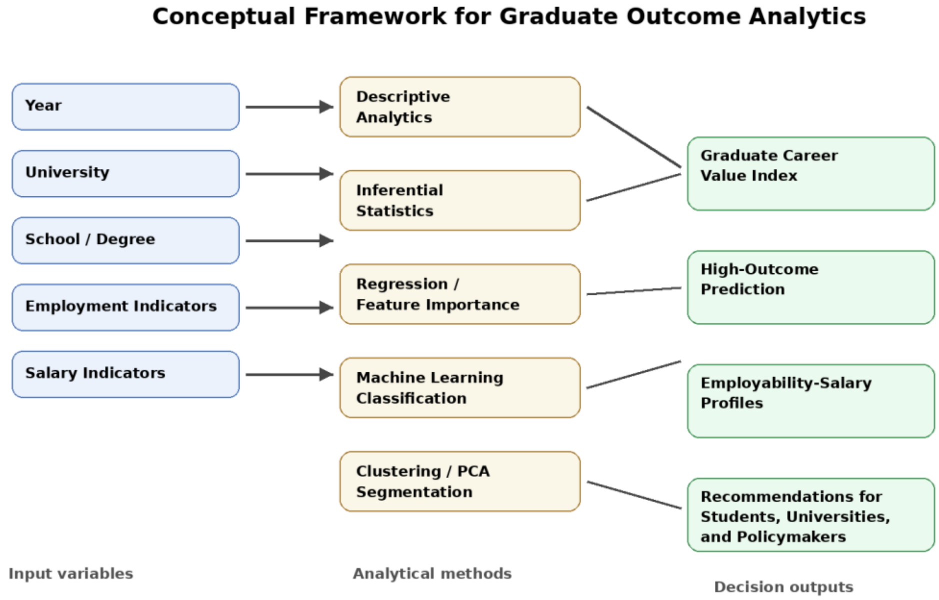

Figure 2. Conceptual framework for graduate outcome analytics.

Graduate Career Value Index

The GCVI will be created as a composite program-level decision-support measure. A starting formulation is: GCVI = 0.30(Overall Employment Score) + 0.30(Full-Time Permanent Employment Score) + 0.25(Median Salary Score) + 0.15(Stability Score). All components will be normalized before aggregation. To address weighting subjectivity, sensitivity analysis will test alternative weighting schemes, and PCA may be explored as a data-driven comparison. The top-quartile threshold for high career value is selected to provide a clear high-outcome category while preserving sufficient class representation for classification modelling.

Solution to RQ1-RQ5

RQ1: How have employment and salary outcomes evolved across Singapore university degree programs from 2013 to 2025?

RQ1 will be addressed using descriptive statistics and trend analysis. Employment and salary indicators, including overall employment rate, full-time permanent employment rate, median gross monthly salary, and median basic monthly salary, will be analyzed across survey years. Mean, median, standard deviation, and percentile summaries will be generated to identify patterns and variations over time. Trend lines, year-over-year changes, compound annual growth rates (CAGR), moving averages, and selected period comparisons will be used to determine whether graduate outcomes have improved, declined, or remained stable during the study period. Where relevant, the analysis will examine the impact of major labor-market events, including the COVID-19 pandemic and subsequent recovery.

RQ2: Do employment and salary outcomes differ significantly across universities and degree programs?

RQ2 will be examined using inferential statistical techniques to determine whether significant differences exist in employment and salary outcomes across universities and degree programs. Analysis of Variance (ANOVA) will be applied when the required assumptions are satisfied. Where normality or homogeneity assumptions are violated, the Kruskal-Wallis test will be used as a non-parametric alternative. Post-hoc tests, including Tukey HSD or Dunn's test, will be conducted to identify specific group differences. Effect sizes, such as eta-squared or epsilon-squared, will also be reported to assess the practical significance of observed differences.

RQ3: Which factors contribute most to graduate salary outcomes and graduate career value across Singapore university programs?

RQ3 focuses on identifying the key drivers of graduate outcomes. Correlation analysis, multiple linear regression, and feature-importance techniques will be used to examine the influence of factors such as university, degree program, year, employment indicators, salary dispersion, and derived performance measures. Gross monthly median salary will serve as the primary salary outcome variable, while the Graduate Career Value Index (GCVI) will be used as an additional composite outcome measure. Regression diagnostics will be performed to evaluate model assumptions, multicollinearity, residual behavior, and overall model fit. Where appropriate, feature-importance measures and optional SHAP-based interpretation may be used to improve model explainability.

RQ4: Can machine learning models accurately predict high-employment, high-salary, and high-career-value degree programs?

RQ4 represents the predictive analytics component of the study. Binary target variables will be created for high-employment, high-salary, and high-career-value programs using predefined percentile thresholds, with the top quartile serving as the initial operational definition of a high outcome. Logistic regression will serve as the baseline classification model, while decision tree, random forest, and optional gradient boosting models will be used to capture more complex relationships. Model performance will be evaluated using train-test splitting and cross-validation. Performance metrics will include accuracy, precision, recall, F1-score, ROC-AUC, balanced accuracy, and confusion matrix analysis. The models will be compared against a baseline classifier to determine whether program-level characteristics can reliably predict high-performing degree programs.

RQ5: Can degree programs be segmented into meaningful employability-salary profiles to support student, university, and policy decision-making?

RQ5 will be addressed using clustering and segmentation techniques. Variables such as employment rates, salary indicators, salary premiums, employment premiums, salary dispersion, and stability scores will be standardized before analysis. K-means clustering will be used as the primary segmentation method, while hierarchical clustering will serve as a robustness check. Principal Component Analysis (PCA) may be used to visualize cluster separation and reduce dimensionality where appropriate. Cluster quality will be assessed using silhouette scores, cluster stability, and practical interpretability. The resulting profiles are expected to identify groups such as high salary-high employment, high employment-moderate salary, high salary-moderate employment, and lower outcome or high-volatility programs, providing actionable insights for students, universities, and policymakers.

Research Hypotheses

RQ	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)
RQ1	Employment and salary outcomes have not changed significantly over time.	Employment and salary outcomes have changed significantly over time.
RQ2	Employment and salary outcomes do not differ significantly across universities and degree programs.	Employment and salary outcomes differ significantly across universities and degree programs.
RQ3	University, degree program, year, and employment indicators are not significantly associated with graduate salary outcomes or graduate career value.	At least one factor is significantly associated with graduate salary outcomes or graduate career value.
RQ4	Machine learning models cannot predict high-outcome degree programs better than a baseline classifier.	Machine learning models can predict high-outcome degree programs better than a baseline classifier.
RQ5	Degree programs cannot be grouped into meaningful employability-salary clusters.	Degree programs can be grouped into meaningful employability-salary clusters.

Evaluation Metrics

Analysis Area	Metric / Test	Formula or Interpretation
Trend	Year-over-year change	$\text{Year-over-year Change} = \frac{Value_t - Value_{t-1}}{Value_{t-1}} \times 100$
Trend	CAGR	$\text{CAGR} = \left(\frac{Ending}{Beginning}\right)^{\frac{1}{n}} - 1$
Group comparison	ANOVA / Kruskal-Wallis	Tests whether group outcomes differ
Regression	R^2, adjusted R^2	Variance explained by the model
Regression	MAE, RMSE	Prediction error for salary outcomes
Classification	Accuracy	$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$

Classification	Precision	$$\text{Precision} = \frac{TP}{TP + FP}$$
Classification	Recall	$$\text{Recall} = \frac{TP}{TP + FN}$$
Classification	F1-score	$$\text{F1-score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$
Classification	ROC-AUC	Class-discrimination ability
Clustering	Silhouette score	Cluster cohesion and separation
Index	Sensitivity analysis	Stability of GCVI rankings under alternative weights

Note. TP = true positive, TN = true negative, FP = false positive, FN = false negative, and GCVI = Graduate Career Value Index.

Recommendation and Application

Target Users

The target users are students, parents, university career services, academic program leaders, policymakers, and workforce planners. Students and parents can use the findings to compare programs across multiple career-outcome dimensions. Universities can identify programs with strong, improving, volatile, or weaker outcomes. Career services can use the segmentation to guide targeted student support. Policymakers can use the results for workforce planning and education-labor-market transparency.

Expected Outputs and Recommendations

Expected outputs include trend dashboards, statistical comparison results, salary and career-value driver analysis, high-outcome prediction models, program clusters, and a concise decision-support summary. The study will not recommend degree choices based only on salary. Instead, it will present multidimensional insights based on employment, salary, stability, relative performance, and program profiles.

Limitations and Ethical Considerations

The study is limited by the aggregated program-level nature of the dataset. It cannot explain individual graduate success and does not include academic performance, internships, socioeconomic background, employer demand, or job-search behavior. Salary values represent early-career outcomes and may not reflect long-term earnings. The GCVI is a constructed analytical aid rather than a definitive ranking of program quality. The study uses public, aggregated data and will avoid stigmatizing specific programs or universities.

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Appendix A: Proposed Analytical Workflow

Stage	Main Activity	Output
1	Download and document GES dataset	Raw data and source note
2	Clean data and create derived variables	Cleaned analysis dataset
3	Perform EDA and trend analysis	Trend charts and summaries
4	Run statistical tests	ANOVA/Kruskal-Wallis results
5	Build regression and GCVI analysis	Driver and career-value results
6	Train classification models	High-outcome prediction metrics
7	Perform clustering and PCA	Employability-salary profiles
8	Prepare dashboard and recommendations	Decision-support output

Appendix B: Proposed Employability-Salary Profiles

Cluster Label	Description	Potential Use
C1: High salary, high employment	Programs with strong salary and strong employment outcomes.	Benchmark programs and high-demand fields.
C2: High employment, moderate salary	Programs with stable employment but moderate salary outcomes.	Career stability guidance and salary growth analysis.
C3: High salary, moderate employment	Programs with strong salary outcomes but weaker employment consistency.	Risk-reward discussion for students and advisors.
C4: Moderate salary, high stability	Programs with consistent outcomes and low volatility.	Reliable career pathway identification.
C5: Lower outcome or high volatility	Programs with weaker or inconsistent early-career outcomes.	Targeted university and career-support intervention.

Appendix C: Proposed GitHub Repository Structure

The proposed submission repository organized as follows:

https://github.com/sundaramkumaran/QM640_GES_Capstone/tree/main

Repository Structure

```
QM640_GES_Capstone/
├── data/
│   ├── raw/
│   │   └── graduate_employment_survey.csv
│   └── processed/
│       └── ges_cleaned.csv
├── notebooks/
│   ├── 01_data_cleaning.ipynb
│   ├── 02_eda_trends.ipynb
│   ├── 03_statistical_tests.ipynb
│   ├── 04_regression_and_gcvi.ipynb
│   ├── 05_machine_learning.ipynb
│   └── 06_clustering.ipynb
├── outputs/
│   ├── figures/
│   └── tables/
├── dashboard/
├── README.md
├── requirements.txt
└── LICENSE
```

Figure 4: Proposed GitHub Repository Folder Tree

Note. The repository structure separates raw data, processed data, analysis notebooks, outputs, dashboard files, and the final report to support reproducibility.